# AFM Lithography Planner: How to Use

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## Overview

The AFM Lithography Planner turns an AFM scan of a graphite flake into one GDS file with the outline of the area to clear and horizontal cutting lines filling it. You load a scan, straighten the flake, mark areas to clear and areas to keep, check the preview, and download the file for the Nanosurf lithography module.

1. Export the scan from Gwyddion as a text file.
2. Open it in the tool and check the scan size.
3. Rotate the image so a flake edge is vertical.
4. Draw clear and keep areas.
5. Check the toolpath, then download the .gds.

## Before you start: export from Gwyddion

Export the scan as an ASCII text file. It keeps the real height values and puts the scan size in the file header, so the tool needs no typing from you.

1. In Gwyddion, select the channel you want (usually Z-Axis forward).
2. Choose File, then Save as, and pick the ASCII matrix (.txt) format.
3. Make sure the file name ends in `.txt`. The tool reads the file type from the extension.

The tool opens only this text format.

## Step 1: Load the scan

Click **Open scan** in the Scan section, or drag the file onto the dark image area. A blue note confirms what loaded, for example "256 × 256 points, 65.00 × 65.00 µm, read from the file header".

- **Scan size** fields: check them against your scan. Edit them only if the file had no size.
- **Level plane**: removes overall tilt. Leave it on.
- **Align rows**: removes horizontal streaks from line-to-line offsets.
- **Flip vertical**: use it if the image appears upside down.

- **Contrast** sliders: set the black and white points. Bad regions, such as a smeared band at the top of a scan, can wash out the flake, so nudge the sliders until the flake edges are clear.
- **Example scan**: loads a made-up scan so you can practise.

The tool reads the scan size from the file, so the Scan size fields normally need no changes.

## Step 2: Rotate the flake

The cutting lines are always horizontal on screen, so rotate the image until the flake sits the way you want the lines to run across it.

**Fastest way: Level on edge.**

1. Click **Level on edge** in the toolbar, or press E.
2. Click two points along a straight flake edge. A length label appears on the line.
3. The image rotates so that edge becomes vertical. The **Level edge to** menu switches this to horizontal.

**Manual rotation.** Use the Rotation slider or number box, the ±0.1° buttons for fine tuning, or the ±90° buttons for quarter turns.

If you already drew shapes, they rotate with the image so they stay on the same part of the flake.

## Step 3: Draw clear and keep areas

Pick the area type at the top left (Clear, Keep or Cut outline), then the tool.

| Control | What it does |
|---|---|
| Clear (red) | Area to be cut away and filled with lines |
| Keep (green, dashed) | Area inside a clear shape that must stay untouched |
| Cut outline (purple) | Draws only the outline of the shape, with no lines inside. Use it to cut out a shape without clearing its inside. |
| Polygon (P) | Click corners; Enter, a double-click, or a click on the first point closes it |

| Control | What it does |
|---|---|
| Rectangle (R) | Drag from one corner to the opposite corner |
| Edit (V) | Click a shape to select it; drag a corner to reshape, or the shape to move it |

**Shift constraints.** Hold Shift while drawing a polygon to force right angles, or while dragging a rectangle to force a square.

**Line lengths.** Every side shows its length in µm as you draw. In Edit mode, the selected shape shows all its side lengths.

**Snap to** rounds corners to a grid (0.1 µm by default). Turn it off for free placement.

**Fixing mistakes.** Backspace removes the last corner while drawing, Esc cancels the shape, and Ctrl+Z undoes. The shape list lets you select a shape, switch it between clear and keep, or delete it.

Keep areas only cut holes in clear areas, and the order you draw the shapes in does not matter.

## Step 4: Check the toolpath

The Pattern section controls the lines and shows what the file will contain.

- **Line spacing**: distance between cutting lines. The default is 0.1 µm. The first line sits half a spacing above the lowest edge of the clear area.
- **Tip speed**: used only for the time estimate. Set it to the speed you run on the AFM.
- **Show toolpath on image**: draws the outline (thick) and lines (thin) over the scan, with a dot where the path starts.
- **Statistics**: line count, line length, outline length, clear area and estimated time.

Zoom in with the scroll wheel to inspect the lines near edges and around keep areas. Hold Space and drag to pan. Lines run in a snake pattern: the bottom row goes left to right, the next right to left, and so on.

## Step 5: Export and load on the AFM

In the Export section:

- **GDS layer** and **datatype**: default to 2 and 0, matching your square.gds. Outlines and lines are written on this one layer.
- **Origin**: Scan center keeps the coordinates relative to the middle of the image. Pattern lower left puts the pattern's corner at 0, 0. You can reposition the pattern in the Nanosurf software either way.
- **Download .gds**: saves `your_file_name.gds`. Type the name you want in **File name** first. The SVG and saved shapes use the same name. Units are 1 nm per database unit, the same as your file.
- **SVG preview**: saves an image of the outline and lines for checking or records.
- **Save shapes** and **Load shapes**: store and reload your drawn shapes as a small JSON file.

In the file, each outline is a closed boundary and each cutting line is a two-point boundary that doubles back on itself, the same pattern your square uses. Copy the .gds to the AFM computer and load it in the lithography module.

**Test before a real cut.** Export a small area first, load it, and confirm that the tip traces the outline and runs the lines horizontally where you expect. If the outline traces wrong, report what the tip does so the outline style can be adjusted.

## Shortcuts

| Key | Action |
|---|---|
| P | Polygon tool |
| R | Rectangle tool |
| V | Edit tool |
| E | Level on edge |
| M | Cycle between clear, keep and cut outline |
| Shift | Right angles (polygon) or square (rectangle) |
| Enter | Close the polygon |
| Esc | Cancel the shape in progress |
| Backspace | Remove the last corner while drawing; delete the selected shape otherwise |

| Key | Action |
|---|---|
| Space + drag | Pan the view |
| Scroll wheel | Zoom |
| Ctrl+Z / Ctrl+Shift+Z | Undo / redo |

## Troubleshooting

| Problem | Fix |
|---|---|
| Text file will not load | Rename it so it ends in `.txt` and check that it holds only numbers after the `#` header lines. |
| Image upside down | Tick **Flip vertical**. |
| Flake barely visible | Adjust the Contrast sliders; turn on **Align rows** if streaks dominate. |
| Pattern size looks wrong | Check the **Scan size** fields against the scan. |
| Download button is greyed out | Draw at least one clear area or cut outline. |
| Message about the outline library | You are offline. Fill lines are still correct; shapes just keep separate outlines. |
| Saved file ends in .gds.txt | Downloaded from the published web page. Rename it to remove .txt, or use the standalone file. |
| Outline traces wrongly on the AFM | Report what the tip does so the outline style can be adjusted. |